7. Project Demonstration Phase

7.1 Introduction

The demonstration phase presents the completed PocketSmart AI application and shows its major functionalities in a real-world usage scenario. It begins with registration and login and continues through expense tracking, budgeting, insights, goals, recommendations, and progress monitoring.

7.2 Objectives of Demonstration

- To demonstrate the overall functionality of PocketSmart AI.
- To show the user authentication system.
- To demonstrate expense tracking and smart categorization.
- To demonstrate AI budget planning, insights, savings goals, and recommendations.
- To display history and the financial dashboard.
- To show the interaction between frontend, backend, database, and Gemini.
- To present the final working application to the evaluator.

7.3 Demonstration Environment

The demonstration is performed in the configured development environment on a laptop with an internet connection, using VS Code to run the FastAPI server, a web browser for the frontend, the configured database, and a valid Google Gemini API key.

7.4 Demonstration Workflow

Step	Feature	Demonstration
1	Registration & Login	Create an account and log in securely.
2	Expense Tracking	Add income and several expenses; show AI-suggested categories.
3	AI Budget Planner	Enter income, fixed costs, and goals to generate a monthly budget.
4	Spending Insights	View the summary of spending patterns and saving opportunities.
5	Savings Goal	Create a goal with a target amount and date and view the saving plan.
6	Recommendations	Enter a need and budget limit to receive budget-friendly options.
7	Finance Chat	Ask a money question and a follow-up question.
8	History	Open previously saved budgets, insights, and chats.
9	Dashboard	View income vs expenses and category charts.
10	Logout	End the session.

7.5 Sample Scenario

A college student with a fixed monthly allowance logs in and records the month's expenses. PocketSmart AI suggests categories, generates a budget, and points out that food delivery

exceeds the planned amount. The student sets a goal to save for a laptop in six months and receives a monthly saving target. When planning a purchase, the student enters a budget limit and receives alternatives with reasons, then asks the chat assistant a follow-up question. The dashboard then shows income against expenses and goal progress.

7.6 Results and Observations

- The application meets the defined functional requirements.
- Gemini provides clear, relevant budget guidance and recommendations.
- The interface is simple and easy to navigate.
- User data is protected through authentication.
- Saved history and the dashboard help users track financial progress.

7.7 Limitations

- AI advice is general guidance, may occasionally be inaccurate, and is not professional financial advice.
- Transactions are entered manually; there is no bank integration.
- An internet connection and Gemini API access are required.
- Response time depends on API availability and network speed.

7.8 Future Enhancements

- Automatic import of bank statements and SMS or receipt scanning.
- Voice input for adding expenses.
- Support for multiple languages and currencies.
- Mobile application.
- Predictive spending alerts and bill reminders.
- Shared family or group budgets.

7.9 Conclusion

PocketSmart AI: Your Smart Budget & Recommendation Assistant combines expense tracking, AI budget planning, personalized recommendations, and progress monitoring in a single application. The demonstration confirms that the system works as intended and gives users a convenient, personalized way to manage their money.